



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Second Year - Semester II Examination – November / December 2023

CHE 2201 –PHYSICAL CHEMISTRY II

Answer all questions

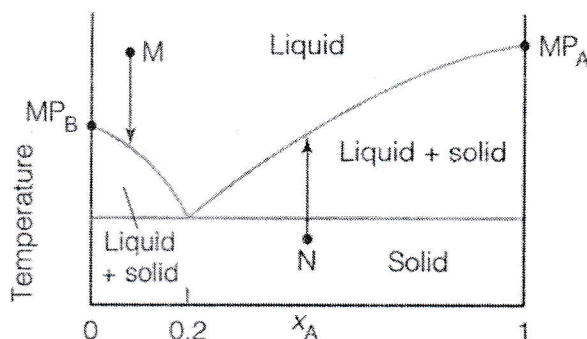
Time: 2 hours

Use of a non-programmable calculator is permitted

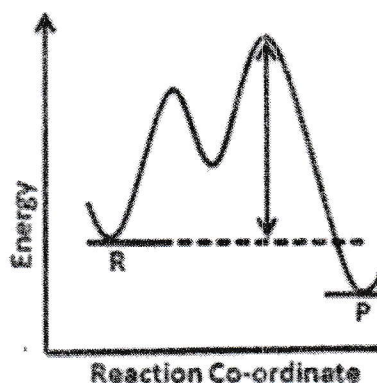
Avogadro's number (N_A) = $6.02 \times 10^{23} \text{ mol}^{-1}$, Gas constant (R) = $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

1

- a) (i) Write an expression for the Gibbs phase rule and define each term in the expression. (20 marks)
- (ii) Calculate the (i) number of components (ii) number of phases and (iii) degrees of freedom of the following systems
 $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, $\text{PCl}_5(\text{S}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ and aqueous solution of sodium acetate (30 marks)
- (iii) At some particular temperature, the vapour pressure of pure benzene, is $2.56 \times 10^4 \text{ N m}^{-2}$ and the vapor pressure of pure toluene, is $9.25 \times 10^3 \text{ N m}^{-2}$. If the mole fraction of toluene in the solution is 0.600 and there is some empty space in the system, Calculate the (i) total vapor pressure in equilibrium with the liquid, (ii) composition of the vapor in terms of mole fraction (20 marks)
- b) The diagram shows a phase diagram of two components, A and B. It also shows two initial points, the dots M and N. Explain the following behavior of (i) the components as the system starts at point M and cools (ii) the behavior of the components as the system starts at point N and warms. (30 marks)



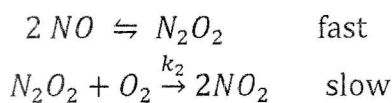
2. a) (i) State the conditions at molecular level that molecules should acquire to initiate a gas phase chemical reaction and (ii) Label and illustrate the following diagram taking into account of an activated complex and a reaction intermediate formed in a given reaction



(30 marks)

- b) Referring to the following reaction, if the reaction proceeds at a rate of $5.00 \text{ mol min}^{-1}$ with respect to NO, Determine the rates with respect to O_2 and NO_2
- $$2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$$
- (20 marks)
- c) Explain the effect of concentration and temperature on reaction rates
- (20 marks)
- d) How would you experimentally determine the order of a reaction by 'isolation method?'
- (30 marks)

3. a) The oxidation of NO to NO_2 occurs via the mechanism given below



Find $\frac{d[\text{NO}_2]}{dt}$ in the presences of large excess of oxygen and show that the reaction order with respect to NO is 2

(30 marks)

- b) Explain the following:

(i) Reaction mechanism (ii) A catalyst can increase the rate of a reaction

(30 marks)

- c) The rate constant determined for the reaction between the hydroxyl radical and chlorobromomethane was found to be $2.09 \times 10^{-13} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at 370 K. Calculate the activation energy of the reaction if the pre-exponential factor is $3.8 \times 10^{-12} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$.
- (20 marks)
- d) With the use of relevant phase diagrams, distinguish between azeotropic and zeotropic distillation

(20 marks)

4. a) Explain the meaning of the following::

- i. Operator
- ii. Born's interpretation of wavefunction

(20 marks)

- b) i. Which of the function e^{ax^2} and $\sin(kx)$ is eigen functions of d/dx and d^2/dx^2 and find their eigen values?
- ii. Calculate the expectation value for position (x) of a wave function between 0 and 2π , if

$$\Psi(x) = \sqrt{\frac{1}{2\pi}} e^{inx}$$

Write the complex conjugate for the wave function and calculate the expectation value

$\langle x \rangle$

(40 marks)

- c) Give the significance of the Hamiltonian operator and write down Schrodinger equation for the motion of a single particle at some positions x , y , and z . Identify all terms there in.

(15 marks)

- d) What is meant by Normalized Wave Function? Probability density of observing a particle at point x is given by:

$$\Psi^2 = A^2 \sin^2\left(\frac{n\pi x}{l}\right)$$

A particle is trapped at ground state in an one-dimensional box with length 'a'. Find the probability of observing this particle at half of the length

(25 marks)